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Adapter Scaling Trade-off and Timestep-Conditioned Scheduling in Multi-view Diffusion Texture Generation

A R T I C L E I N F O

Keywords: multi-view diffusion model, 3D texture generation, adapter scaling, timestep-conditioned scaling, shape-texture trade-off, geometric control, texture quality assessment

A B S T R A C T

Multi-view diffusion pipelines for 3D asset texturing often inject geometric conditions, such as normal and depth maps, through adapter modules. However, the adapter scale used at inference time is usually chosen empirically, although the generated views are later projected and baked into the final mesh texture. This paper studies how adapter strength affects geometry-conditioned multi-view texture generation. We observe a shape-texture trade-off: larger scales improve structural metrics such as FG-SSIM, but can weaken coherent color variation, flatten local surface appearance, and amplify over-controlled high-frequency artifacts. We identify this as an adapter-specific temporal conflict, where geometric residuals are most useful during mid-stage structure refinement, while excessive residual injection in early or late denoising can disturb global formation or suppress texture synthesis. Based on this observation, we propose Timestep-Conditioned Adapter Scaling (TCAS), a training-free inference strategy that applies strong adapter correction only in the middle denoising stage and uses conservative scales elsewhere. The selected C3 schedule is chosen on a 24-object probe set from 17 variants and then transferred unchanged to a 300-object validation set. Experiments show that C3 achieves FG-SSIM comparable to aggressive uniform scaling while improving PSNR by 0.96 dB and retaining more foreground texture variation. CLIP-IQA, local visual audits, and preference evaluation further support that TCAS provides a better balance between geometric adherence and texture preservation. A supporting Full Adaptive Correction experiment suggests that the same stage-wise principle can be extended to learned residual modulation, while TCAS remains the main plug-and-play strategy.

1. Introduction

Recent advances in single-image 3D generation have substantially improved the recovery of geometry, multi-view consistency, and sparse-view reconstruction [1, 2, 3, 4, 5, 6, 7, 8]. As geometric generation models become increasingly mature, generating high-quality textures on an existing 3D shape has become a key factor affecting the realism and usability of 3D content. For an object whose geometry has already been determined, the final visual quality depends not only on whether the shape is complete, but also on whether the texture is faithful to the reference image, aligned with the 3D surface, and rich in local details.

Existing multi-view texture generation methods usually generate RGB images from multiple target views using 2D image diffusion models, and then map the generated images back to the 3D surface through projection or texture baking [9, 10, 11, 12]. This route benefits from the generative capability of image diffusion models, but it still faces challenges in reference appearance alignment, geometry-texture consistency, and local texture detail preservation. Geometry-controlled multi-view diffusion methods represented by MVPainter introduce geometric conditions such as normals and depth maps to improve the alignment between generated textures and the underlying 3D surface, demonstrating the importance of explicit geometric control for 3D texture generation [13].

This problem is practical for graphics asset production because the generated multi-view images are not only evaluation frames: they are projected and baked into the final mesh texture. Local color collapse, spurious high-frequency marks, or view-dependent over-control can therefore become persistent defects in the textured asset.

However, introducing geometric conditions is not only a question of whether a control signal should be used, but also a question of how strongly it should be applied. In adapter-augmented diffusion pipelines, geometric conditions are usually injected into the model through an adapter or control branch, and their influence is regulated by a scaling factor. In practice, this factor is often set empirically: a small scale may lead to insufficient geometric constraints, whereas a larger scale is usually expected to improve structural consistency.

In multi-view texture generation, however, stronger adapter scaling is not necessarily better. Texture generation requires not only contours and structures to be consistent with geometry, but also surface color variations, material details, and high-frequency patterns to be preserved. Overly strong geometric control may force the model to follow normal, depth, or edge conditions too aggressively, thereby suppressing the intrinsic texture synthesis capability of the base diffusion model. In this case, some structural metrics may still improve, while the visual result suffers from texture flattening, reduced color variation, and detail loss.

Therefore, adapter scaling in geometry-controlled multi-view diffusion deserves further analysis. Compared with prior work that mainly studies how to build texture generation pipelines, how to introduce geometric conditions, and how to improve overall generation quality, adapter strength is a fine-grained inference-time control problem. It directly affects the balance between shape consistency and texture detail, and is an easily overlooked factor in multi-view texture generation quality.

This problem is related to, but not solved by, existing guidance scheduling strategies. CFG scheduling changes the extrapolation between conditional and unconditional predictions, while a geometry adapter injects residuals into intermediate representations. Generic ControlNet or adapter scale scheduling similarly treats condition strength mainly as a condition-adherence knob. In texture generation, however, the key question is not only how strongly to follow the geometry, but also when geometric residuals should dominate and when they should recede so that the base model can synthesize color, material, and high-frequency texture. Simple monotonic schedules such as linear or cosine decay do not explicitly address this non-monotonic requirement.

The goal of this paper is not to propose a new full texture generation system, but to analyze and improve inference-time adapter scaling within a fixed geometry-controlled multi-view diffusion pipeline.

The main contributions of this paper are as follows:

(1) We systematically diagnose the shape-texture trade-off caused by adapter scaling in multi-view diffusion texture generation, showing that higher scales can improve structural metrics while simultaneously weakening coherent color variation, flattening surfaces, and reducing foreground high-frequency response.

(2) We identify an adapter-specific temporal conflict between geometric residual injection and texture formation, and propose Timestep-Conditioned Adapter Scaling (TCAS), which concentrates strong geometric correction in the middle denoising stage while using conservative scales in the early and late stages.

(3) Through uniform scale sweeps, manual local texture audits, a 24-object probe over 17 variants, and a 300-object validation without re-searching, we show why fixed high scales and direct guidance-schedule transfers are insufficient.

(4) We provide large-scale texture-diagnostic, CLIP-IQA, and blinded preference analyses showing that TCAS reduces the foreground-variation and perceptual-quality loss caused by full high-scale scaling.

As supporting evidence rather than a separate main method, we also report a lightweight Full Adaptive Correction (FAC) experiment, which shows that the stage-wise adapter-control principle can be instantiated with learned temporal, spatial, and frequency-aware residual modulation.

2. Related Work

2.1. 3D Texture Generation and Multi-view Diffusion

Single-image 3D generation is often decomposed into geometry and texture generation. Single-image-to-3D, multi-view diffusion, and sparse-view reconstruction methods have improved structural stability and shape completeness [1, 2, 3, 4, 5, 6, 7, 8], but high-quality texture remains a bottleneck. Existing texture methods typically synthesize multi-view RGB images with 2D diffusion models and then project or bake them onto the 3D surface [9, 10, 11, 12], which still leaves reference appearance misalignment, geometry-texture inconsistency, and local detail loss.

Methods such as MVPainter formulate 3D texture generation as geometry-controlled multi-view diffusion [13]; MV-Adapter further shows that adapter-style multi-view control can improve cross-view consistency without fully retraining the base model [14]. These works inject normals, depth maps, or related geometry signals to align textures with the surface, but they mainly study how to introduce geometry rather than how strongly it should act during inference.

2.2. Adapter-augmented Diffusion Models and Geometric Control

Adapter modules have become an important mechanism for introducing additional conditions into diffusion models. ControlNet injects spatial conditions such as edges, depth maps, and normals into a pretrained diffusion model through a parallel branch [15]. IP-Adapter introduces image prompts through decoupled attention [16]. LoRA and its variants implement lightweight model adaptation through low-rank residual updates [17, 18, 19]. Although these methods differ in architecture, they generally include some form of scaling factor that controls the influence of the external condition or adapter residual on the base model.

In practice, the adapter scale is often treated as an empirical hyperparameter. A larger scale improves condition adherence, but in texture generation it may weaken the base model's ability to synthesize color, material, and high-frequency detail. Adapter scaling is therefore a balance between geometric constraints and texture fidelity: given a fixed adapter and fixed geometric conditions, how should the strength be scheduled to preserve both?

2.3. Diffusion Guidance Scheduling and Timestep Control

The denoising process of diffusion models is highly stage-dependent. Early steps mainly determine global layout and coarse structure, middle steps refine shape and semantic structure, and late steps are more responsible for color, texture, and high-frequency details [20, 21]. Based on this observation, many guidance scheduling strategies have been explored in both research and practice. For example, classifier-free guidance scales may be adjusted linearly, with cosine schedules, or dynamically to mitigate oversaturation, detail loss, or reduced diversity caused by overly strong guidance [22, 23]. In addition, the development of latent diffusion, high-resolution diffusion, and Transformer-based diffusion models further shows the close relationship among generation quality, noise schedules, conditional injection, and network architecture [24, 25, 26, 27, 28, 29, 30, 31].

These methods all exploit the staged nature of denoising to regulate conditional strength. However, different control signals act through different mechanisms. CFG scheduling adjusts the weight between conditional and unconditional predictions, mainly affecting the balance among semantic control, diversity, and visual fidelity. By contrast, the scale of a geometric adapter acts directly on intermediate features or residual branches, controlling how strongly geometric conditions intervene in the generation process. ControlNet-style scale control is closer in spirit, but it is still usually interpreted as a condition-adherence strength; directly applying a larger, smaller, or smoothly decayed control scale does not specify which denoising stage should receive strong geometric residuals for texture generation.

This distinction matters because the desired schedule for geometry-controlled texture generation is not necessarily monotonic. Decay or warm-up can reduce over-control in one part of the trajectory, but cannot simultaneously avoid unstable early residuals, strengthen mid-stage geometry refinement, and weaken late residuals that compete with color and high-frequency texture formation. Adapter scheduling therefore requires a dedicated analysis of geometric residual injection rather than a direct transfer of CFG or generic ControlNet scale scheduling.

3. Method

3.1. Task Definition and Base Pipeline

This paper builds on the MVPainter-style geometry-controlled multi-view texture generation framework [13] and studies how the inference-time strength of geometric condition injection affects the generated results. Given a single reference image and the corresponding 3D geometry, the base pipeline first generates RGB texture images from six target views and then obtains a textured mesh through projection or texture baking. We do not modify the base UNet, VAE, or pretrained adapter weights. Instead, the adapter scaling factor is treated as the key inference-time control variable.

During multi-view diffusion, the 3D geometry is rendered into normal maps, depth maps, and foreground masks, denoted as:

$$G = \{G_{\mathrm{normal}}, G_{\mathrm{depth}}, G_{\mathrm{mask}}\}. \tag{1}$$

The geometric adapter maps these conditions to a residual correction term and injects it into the intermediate hidden states of the UNet. For the t-th denoising step, the adapter injection can be written as:

$$h_t' = h_t + s(p_t) \cdot A_\phi(h_t, G), \tag{2}$$

where h_t and h_t' denote the hidden states before and after injection, p_t denotes the sampling progress, and $s(p_t)$ is the adapter scaling strength at the current denoising stage. This formulation highlights a key difference from CFG-style guidance: changing $s(p_t)$ does not reweight two predicted scores, but changes the magnitude of a geometry-conditioned residual added to the feature stream. A larger s usually enhances geometric condition adherence. However, if strong residuals are continuously applied in the late denoising stage, they may suppress the synthesis of color, material, and high-frequency details by the base diffusion model. This paper therefore studies how to schedule s during denoising to balance shape consistency and texture detail preservation.

3.2. Shape-Texture Diagnostic Metrics

Adapter scaling cannot be evaluated by a single structural metric. We use shape and structure metrics based on SSIM [32] (FG-SSIM, Edge-SSIM, Crop-SSIM), signal/perceptual metrics (PSNR and LPIPS [33]), and texture diagnostics (Laplacian variance, RGB standard deviation, gradient magnitude) to detect weakened color variation, surface smoothing, and high-frequency response loss.

Laplacian variance, RGB standard deviation, and gradient magnitude are not standalone perceptual texture-quality metrics: larger values can also arise from noise, incorrect patterns, or artifacts. We use them only as

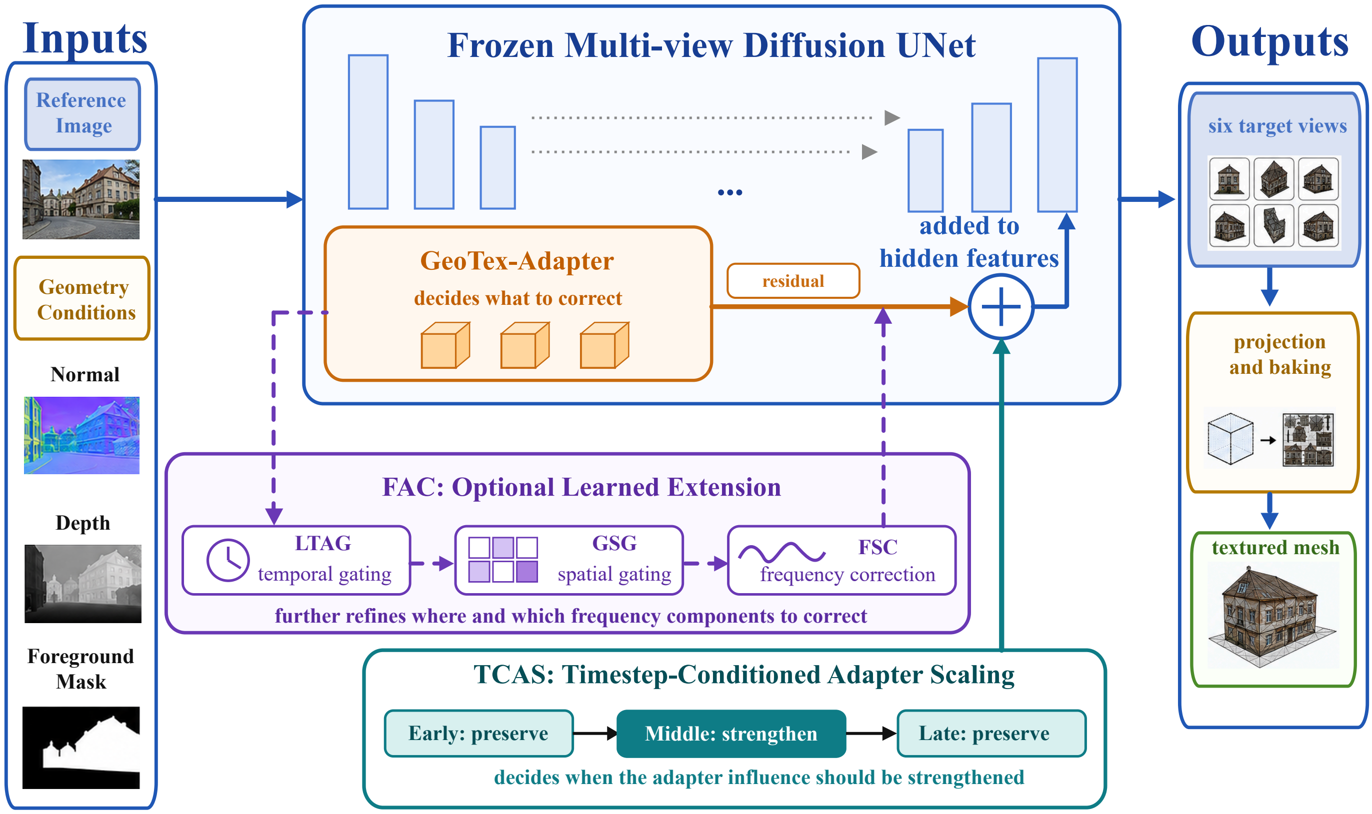

Fig. 1. Method overview. GeoTex-Adapter generates geometric residuals and injects them into UNet features. TCAS dynamically adjusts the residual strength according to denoising stages to balance geometric consistency and texture fidelity. FAC is an optional extension module that further refines residual modulation along temporal, spatial, and frequency dimensions. The final multi-view outputs are projected and baked into a textured 3D mesh.

proxy indicators for the failure mode observed here, where high adapter scales make surfaces overly uniform and suppress local variation. Texture conclusions therefore rely on paired comparisons under identical views and masks, agreement across diagnostics, and local visual audits checking whether retained variation is coherent appearance rather than random artifacts.

Taking Laplacian variance as an example, the foreground texture response is defined as:

$$M_{\text{tex}} = \text{Var}(\text{Lap}(I_{\text{out}} \odot G_{\text{mask}})), \tag{3}$$

where Lap denotes the Laplacian filter, G_{mask} denotes the foreground mask, and Var denotes spatial variance. A larger M_{tex} usually corresponds to stronger local high-frequency variation, but it is not necessarily equivalent to better texture. In this paper, the more reliable signal is the opposite pattern: if structural metrics improve while Laplacian variance, RGB standard deviation, or gradient magnitude consistently decrease and visual crops show smoother surfaces, the improvement may mainly come from contour smoothing or boundary convergence rather than true texture quality improvement.

3.3. Timestep-Conditioned Adapter Scaling

The diffusion denoising process is stage-dependent: early steps mainly establish the global layout and coarse contours; middle steps refine geometric structure and cross-view alignment; late steps are more responsible for color, material, and local texture synthesis. Based on this observation, we propose Timestep-Conditioned Adapter Scaling (TCAS), which replaces a globally fixed adapter scale with a stage-dependent scale function. TCAS is not intended as a generic smooth guidance schedule. Its low-high-low form encodes the adapter-specific hypothesis that geometry residuals should be strongest only after coarse structure becomes stable and before late texture synthesis dominates.

To avoid ambiguity caused by different raw timestep directions in different schedulers, we use sampling progress p to represent the denoising process, where $p = 0$ denotes the start from noise and $p = 1$ denotes the end of sampling close to the final image state. TCAS divides denoising into early, middle, and late stages:

$$s(p) = \begin{cases} s_e, & 0 \leq p < 1/3 \\ s_m, & 1/3 \leq p < 2/3 \\ s_l, & 2/3 \leq p \leq 1 \end{cases}. \tag{4}$$

The selected C3 configuration used in this paper is:

$$(s_e, s_m, s_l) = (1.25, 2.50, 1.25). \tag{5}$$

This design follows three principles. First, the early stage uses a conservative scale to avoid disrupting global structure formation with strong geometric residuals. Second, the middle stage uses a higher scale to strengthen geometric constraints from normals, depth maps, and foreground contours. Third, the late stage reduces the scale again to prevent geometric residuals from suppressing color variation, material detail, and high-frequency texture synthesis. This non-monotonic schedule is therefore different from simply decaying or increasing the adapter strength across all timesteps. TCAS introduces no new parameters and does not modify the base model or adapter weights; it only replaces the scaling factor during inference, so its computational overhead is negligible.

3.4. From Fixed Scheduling to Adaptive Correction: The FAC Framework

GeoTex-Adapter, TCAS, and FAC correspond to three dimensions of adapter control. GeoTex-Adapter determines what geometric residual is injected. TCAS provides a training-free temporal rule: strengthen residuals mainly in the middle geometry-refinement stage and weaken them during early formation and late texture synthesis. FAC develops this principle into lightweight temporal, spatial, and frequency modulation.

GeoTex-Adapter is used as a fixed geometric residual backbone. It receives a 5-channel geometric input consisting of a 3-channel normal map, a 1-channel depth map, and a 1-channel foreground mask. A multi-scale geometric encoder produces a feature pyramid, and residual corrections are injected into multiple resolution levels of the UNet. TCAS acts on the global temporal scale of these residuals to control the influence of geometric conditions at different denoising stages without changing the adapter architecture.

To further examine whether stage-wise adapter control can be extended to learned adaptive correction, we introduce Full Adaptive Correction (FAC). FAC contains three lightweight modules: LTAG learns timestep-dependent gating, GSG generates spatial gates from geometric features, and FSC performs frequency-selective correction on adapter residuals. Unlike fixed TCAS, FAC finely modulates adapter residuals in the temporal, spatial, and frequency domains. In the FAC experiments, TCAS provides the foundational temporal schedule, while GSG and FSC extend it to spatial and frequency-aware residual correction.

FAC requires learning a small number of additional parameters, so TCAS remains the training-free main strategy. FAC serves as its learned adaptive instantiation and shows that stage-wise adapter control can be generalized to finer-grained, content-adaptive residual modulation.

4. Experiments

4.1. Experimental Setup

Experiments are conducted using an MVPainter-style geometry-controlled multi-view diffusion pipeline. Each object contains a single reference image and corresponding 3D geometry. During preprocessing, the 3D geometry is rendered into target-view normal maps, depth maps, foreground masks, and ground-truth RGB images. The model generates six target views at a time, and all comparisons with ground-truth renderings are performed under unified views, resolution, and foreground masks.

Except for the FAC framework experiment, all scaling experiments keep the object list, reference image, target views, random seed, base UNet, VAE, adapter weights, and number of denoising steps fixed; only the adapter scale or schedule is changed. The number of sampling steps is 50. C3 is selected using only a 24-object probe set; the same schedule is then frozen and applied to the 300-object validation set, where no parameter search is performed. This probe-to-validation protocol isolates the effect of adapter scaling from changes in model architecture, training data, sampling budget, or validation-set tuning.

Accordingly, the comparisons are among inference-time adapter scaling strategies under the same base pipeline, not among different texture generation architectures. Uniform scales serve as deployment baselines, while probe variants test whether the trade-off can be resolved by globally high scales, layer-wise redistribution, or moving high strength to early, late, or multiple denoising stages.

We use four evaluation sets: (1) a 50-object scale-sweep set for analyzing the overall trend of structural metrics under 13 uniform scales; (2) a 26-object texture audit set for comparing texture degradation under $s = 1.25$, $s = 2.25$, and $s = 2.50$; (3) a 24-object probe set for evaluating 17 uniform, layer-wise, and timestep-conditioned scheduling variants and selecting C3; and (4) a 300-object validation set for testing the selected C3 schedule, uniform baselines, and the FAC extension without re-searching TCAS.

Metrics include shape/structure (FG-SSIM, Edge-SSIM, Crop-SSIM), signal/perceptual fidelity (PSNR, LPIPS), and texture diagnostics (Laplacian variance, RGB standard deviation, gradient magnitude). Metrics are averaged over six target views and objects; texture conclusions additionally use object-level win rates, significance tests, and local audits to avoid relying on a single structural or high-frequency statistic.

FAC uses the same object list and protocol as the 300-object validation. TCAS is the baseline; LTAG, LTAG+GSG, and Full FAC are ablations. The base model and original adapter remain frozen, and only lightweight correction parameters are learned.

4.2. Shape-Texture Trade-off of Adapter Scaling

Table 1 reports the results of a 13-scale uniform adapter sweep on 50 objects.

As the scale increases from $s = 0.75$ to $s = 2.50$, ΔFG-SSIM rises from -0.013 to $+0.145$, and the number of objects with positive improvement increases from 20/50 to 44/50. This indicates that stronger adapter residuals indeed improve the model's adherence to geometric conditions such as normals, depth maps, and foreground contours. Figure 2 visualizes this trend: FG-SSIM keeps in-

Table 1. Uniform adapter scale sweep results.

Scale	ΔFG-SSIM↑	POS	ΔCrop-SSIM↑	ΔPSNR↑
0.75	−0.013	20/50	+0.113	7.53
0.90	+0.024	30/50	+0.137	8.32
1.00	+0.047	34/50	+0.147	8.79
1.10	+0.070	36/50	+0.159	9.08
1.15	+0.071	36/50	+0.157	9.25
1.25	+0.088	37/50	+0.161	9.37
1.35	+0.100	39/50	+0.163	9.63
1.50	+0.113	40/50	+0.163	9.76
1.60	+0.120	42/50	+0.166	9.83
1.75	+0.127	43/50	+0.173	**9.97**
2.00	+0.134	**44/50**	+0.169	9.96
2.25	+0.142	43/50	**+0.175**	9.91
2.50	**+0.145**	**44/50**	+0.165	9.91

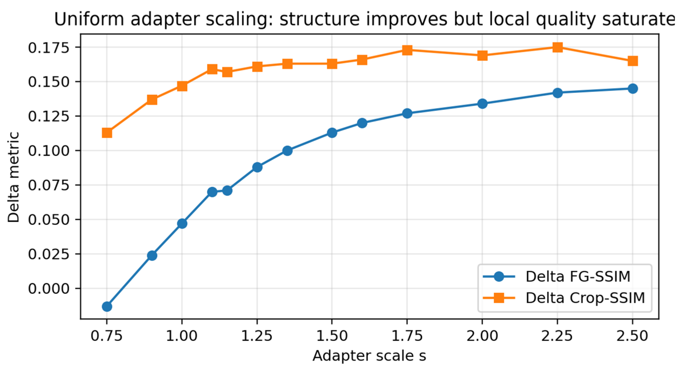

Fig. 2. Structural metric changes under uniform adapter scaling. The curves show that FG-SSIM keeps increasing with adapter scale, while Crop-SSIM saturates and slightly decreases at high scales. Stronger geometric control therefore mainly improves structural alignment, but does not necessarily keep improving local foreground quality.

creasing with adapter scale, while Crop-SSIM saturates and slightly decreases at high scales.

Overall, the FG-SSIM improvement is stable, but the marginal benefit becomes smaller in the high-scale region. For example, increasing the scale from $s = 2.25$ to $s = 2.50$ yields only a limited FG-SSIM gain, while Crop-SSIM already decreases. This suggests that the nature of the high-scale benefit may change: the gain comes more from contour smoothing and foreground alignment than from preservation of local texture details. Figure 3 shows the per-object ΔFG-SSIM heatmap, confirming that most objects gain higher structural scores as the scale increases, but with clear object-level differences.

4.3. Texture Audit: High Scale Causes Texture Flattening

To determine whether the FG-SSIM gain at high scales reflects better visual quality, we conduct a manual local texture audit on 26 objects, comparing $s = 2.50$ with $s = 1.25$. Figure 4 is a local texture comparison: the key evidence is the progressive loss of coherent color variation, material marks, and high-frequency patterns as the scale increases. Retained high-frequency response is counted as

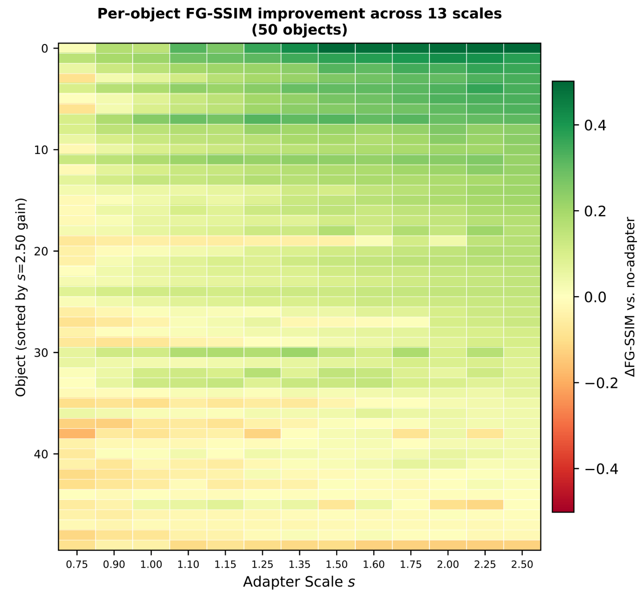

Fig. 3. ΔFG-SSIM heatmap for 50 objects under 13 adapter scales. Most objects obtain higher structural gains as the scale increases, but there are clear object-level differences.

texture preservation only when it corresponds to coherent appearance rather than noise or obvious artifacts.

The results show that 24 out of 26 objects (92%) exhibit "texture loss without true shape gain." Here, true shape gain means visible improvement in effective foreground area, contour structure, or geometric detail, not merely a higher FG-SSIM from smoothed boundaries or averaged contour errors. Thus, high adapter scales can improve structural metrics without improving visual shape or texture quality. Figure 5 quantifies this texture loss.

Overly strong geometric control can suppress color and detail synthesis during late denoising, making surfaces more uniform. Scale selection should therefore not rely only on FG-SSIM or PSNR; texture diagnostics are useful for detecting flattening, but must be interpreted with local visual inspection because high-frequency statistics alone cannot distinguish material detail from noise or artifacts.

4.4. Timestep-Conditioned Adapter Scaling: Shape-Texture Balance of C3

Based on the texture audit, we evaluate 17 adapter scaling variants on the 24-object probe set and freeze the selected schedule before 300-object validation. The variants include uniform baselines, layer-wise scaling, and timestep-conditioned schedules with high strength in early, middle, late, or broader temporal regions. This rules out the simple explanation that any stronger control scale or guidance-like schedule solves the problem.

All variants are evaluated using ΔFG-SSIM for shape gain and ΔLap Var for texture response, computed against ground-truth foreground renderings under the same views and masks. The probe objective is not the highest structural score, but a stable compromise that avoids the trap of

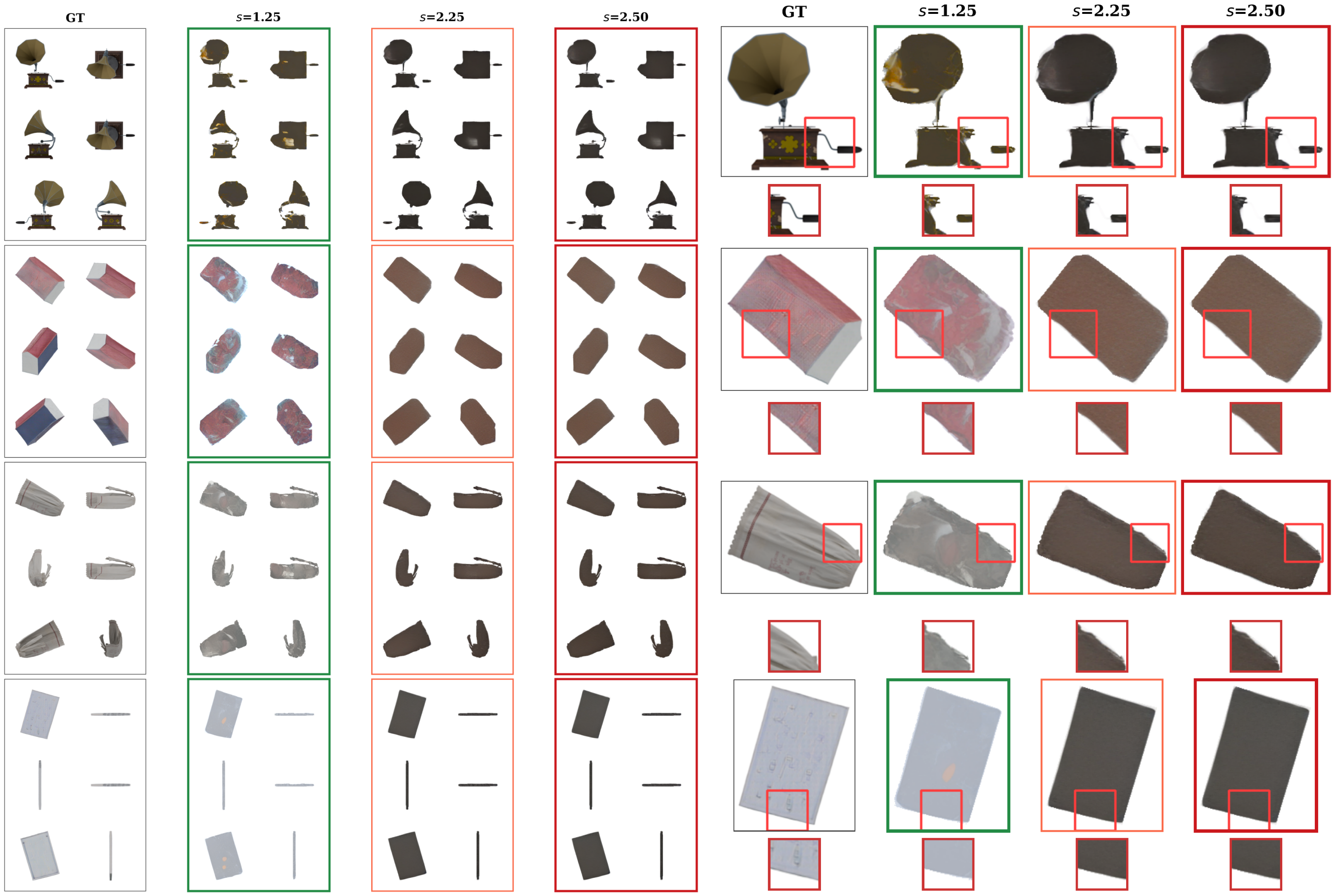

Fig. 4. Texture degradation under different adapter scales. (a) Multi-view texture comparison. Each object is shown from six target views. Compared with conservative scaling $s = 1.25$, higher scales $s = 2.25$ and $s = 2.50$ show more obvious color convergence and surface smoothing in multiple views. (b) Local zoom-in comparison. Red boxes mark local texture regions; green borders denote conservative scaling $s = 1.25$, and red borders denote aggressive scaling $s = 2.50$. Within each row, the intended comparison is the progression from $s = 1.25$ to higher scales: local color variation, material details, and high-frequency texture gradually weaken, and the surface becomes more uniform.

Table 2. Shape-texture trade-off on the 24-object probe set. Delta texture metrics are computed relative to the corresponding ground-truth foreground renderings under the same views and masks; ΔLap Var closer to 0 indicates smaller loss of foreground high-frequency response, which is interpreted together with visual texture audits.

Variant	Category	ΔFG-SSIM↑	ΔLap Var → 0	Assessment
C4	Late-stage high	**0.105**	−0.0120	Trap
C2	Timestep sched.	0.104	−0.0110	Trap
B2	Layer-wise sched.	0.090	−0.0135	Trap
A_s2.00	Uniform	0.087	−0.0062	Trap
C3 (TCAS)	Mid-stage high	0.084	**−0.0006**	Promising
A_s2.25	Uniform	0.079	−0.0096	Trap
A_s2.50	Uniform	0.078	−0.0108	Trap
B6	Layer-wise sched.	0.000	−0.0027	OK

"higher structural metrics but flattened texture." A schedule that improves FG-SSIM through late strong residuals while reducing texture response does not solve the adapter-specific conflict.

To test whether generic CFG or ControlNet-style scheduling can solve the trade-off, we evaluate linear decay, cosine decay, linear warm-up, and a smooth cosine bump on the same 24-object probe set, with fixed low/high endpoints and identical pipeline, objects, seed, 50 denoising steps, and metrics.

Table 3 shows that common guidance schedules do not solve the problem. Decay schedules start with strong early residuals and still retain high late strength, producing unstable outputs and lower PSNR. Warm-up avoids early over-control but peaks in the late texture-synthesis stage. The cosine bump is closest to C3, yet its effective high-scale duration is shorter than the full middle third used by C3. C3 therefore obtains the highest PSNR and higher FG-SSIM than monotonic decay and smooth bump schedules, with FG-SSIM win rates of 100% over fixed high, linear decay, and cosine decay, 96% over cosine bump, and 67% over linear warm-up. Although fixed-low scaling has a higher probe FG-SSIM than C3, it is included only as a conservative reference: on the 300-object validation set, its FG-SSIM and Edge-SSIM are clearly lower than C3 (Table 5), indicating insufficient geometric correction. The goal is therefore not to beat fixed low on every probe metric, but to obtain a schedule that transfers with better geometry-texture balance. The probe diagnostics are used

Table 3. Comparison with generic guidance-style schedules on the 24-object probe set. The table reports structural and signal-fidelity metrics together with diagnostic assessment. Probe-set Laplacian/RGB diagnostic ratios are not used as texture-quality scores here because extremely large values can be dominated by unstable color or noise amplification; they should not be compared with the retention ratios in Table 7.

Schedule	Form	FG-SSIM↑	PSNR↑	Probe diagnostic outcome	Assessment
Fixed low	1.25 uniform	**0.3109**	16.13	Conservative variation	Reference
Fixed high	2.50 uniform	0.1900	15.52	Artifact amplification	Over-controlled/artifact-prone
Linear decay	2.50 → 1.25	0.2110	15.58	Artifact amplification	Unstable geometry/artifacts
Cosine decay	2.50 → 1.25	0.2081	15.48	Artifact amplification	Unstable geometry/artifacts
Linear warm-up	1.25 → 2.50	0.2942	16.31	Late-stage over-control	Over-constrained texture stage
Cosine bump	1.25 → 2.50 → 1.25	0.2554	16.10	Insufficient middle concentration	Weak geometry
C3 (TCAS)	1.25 → 2.50 → 1.25 piecewise	0.2971	**16.67**	Stable diagnostics	Best trade-off

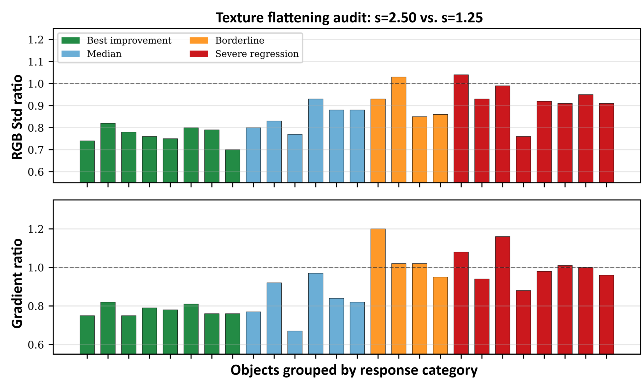

Fig. 5. Texture flattening at $s = 2.50$. Top: RGB standard deviation ratio on 26 objects ($s = 2.50$ / $s = 1.25$). Bottom: gradient magnitude ratio. Values below 1.0 indicate reduced foreground variation. Together with the local visual audit in Figure 4, these reductions support the observation of texture flattening. Even the four best-improvement objects (green) show 24% to 30% lower diagnostic responses despite positive FG-SSIM gains.

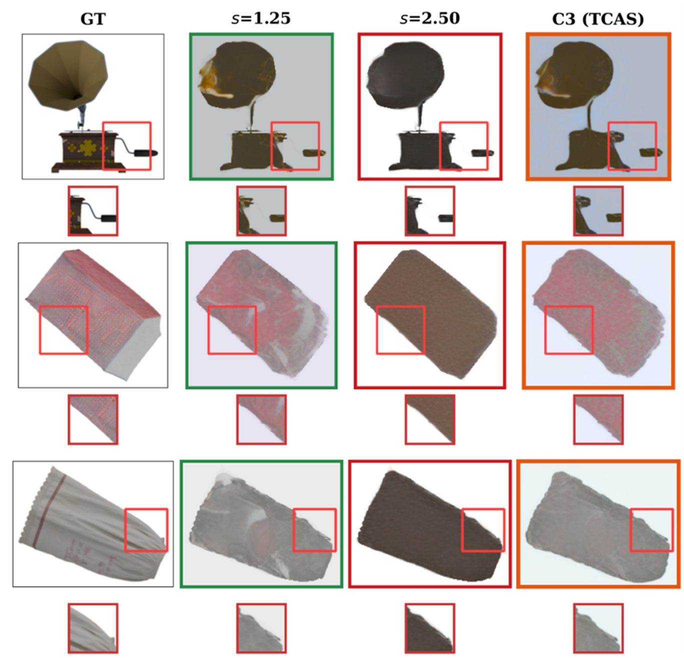

Fig. 6. Qualitative comparison between C3 and uniform adapter scaling. Columns show GT, conservative uniform scaling $s = 1.25$, aggressive uniform scaling $s = 2.50$, and C3 (TCAS). Compared with $s = 2.50$, C3 retains more local color variation and material texture in the red-boxed zoom-in regions while preserving stronger geometric correction than the conservative scale. Orange borders denote C3 outputs, and red boxes indicate zoom-in texture regions.

only to flag unstable amplification or over-control, while the larger 300-object study reports texture retention with a consistent $s = 1.25$ reference.

Table 2 summarizes representative variants. C4 and other late-stage high-scale strategies obtain higher ΔFG-SSIM, but with clear Laplacian-variance drops, showing that structural gains come with texture loss. Uniform high-scale strategies show the same issue: $s = 2.50$ reaches +0.078 ΔFG-SSIM but −0.0108 ΔLap Var. C3 reaches +0.084 ΔFG-SSIM with only −0.0006 ΔLap Var, close to zero texture loss, and is frozen before large-scale validation.

The C3 configuration uses $s = 1.25$ in early/late stages and $s = 2.50$ in the middle stage. This follows denoising roles: early latents are noisy, middle signals support geometry refinement, and late steps synthesize color, material, and fine texture, where adapter intervention should recede.

Figure 6 provides a qualitative comparison between C3 and uniform adapter scaling. The red-boxed zoom-in regions make the intended comparison visible: conservative scaling preserves more color variation but provides weaker structural correction, whereas aggressive scaling strengthens geometric constraints but flattens surface texture. C3 applies strong geometric correction in the middle stage and reduces adapter intervention in the late stage, thereby preserving more local texture than $s = 2.50$ while retaining stronger geometric correction than the conservative scale.

Figure 7 shows the 17 variants in shape-texture space. Most high-shape-gain variants fall into the Trap region with large texture loss, whereas C3 keeps relatively high shape gain and near-zero texture loss. Its value is not maximizing every metric, but avoiding "higher structural scores but flattened texture."

Mechanistically, C3 is tied to denoising-stage behavior rather than validation-set tuning or a generic monotonic rule. Strong early residuals are unstable under high noise, middle-stage signals make normals/depth/contours most effective, and late strong residuals compete with color, material, and fine texture synthesis.

This matches the variant results: early high-scale vari-

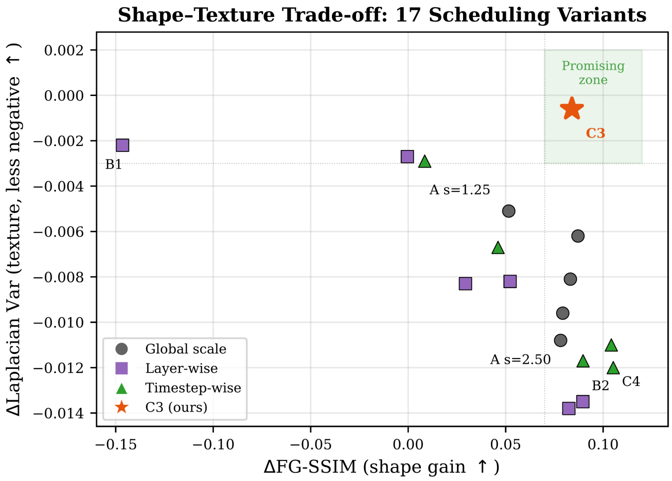

Fig. 7. Shape-texture Pareto trade-off among 17 variants. The horizontal axis is shape gain, where farther right is better; therefore, the bottom-right region represents a more desirable compromise. C3 (star) lies in the Promising region, with high shape gain and nearly zero texture loss. Most high-shape-gain variants fall into the Trap region, where structural scores are obtained at the cost of large texture loss. Variants such as B6 have smaller texture degradation but almost no shape gain.

ants give limited structural gain, C4 obtains a high structural score but large texture loss, uniform high scales improve condition adherence while losing detail, and an aggressive C3 variant reduces signal fidelity. Stronger ControlNet-like scaling alone therefore does not resolve the conflict.

Table 4. Comparison of representative variants under extended texture metrics. All delta texture metrics are computed relative to the corresponding ground-truth foreground renderings under the same views and masks. This table verifies that the advantage of C3 is not due to a single Laplacian variance metric.

Variant	ΔLap Var	ΔRGB Std	ΔGrad	ΔHF Energy
C3	**−0.001**	**−0.023**	**+0.142**	+0.058
C4	−0.012	−0.032	+0.086	+0.062
A_s2.50	−0.011	−0.044	+0.054	**+0.078**
A_s2.00	−0.006	−0.040	+0.100	+0.070
A_s1.25	−0.005	−0.032	+0.104	+0.053

C3 has nearly zero loss in Laplacian variance and remains relatively stable in RGB standard deviation, gradient magnitude, and high-frequency energy. Compared with C4 and uniform high-scale strategies, C3 shows weaker texture degradation, indicating that its shape-texture balance is not an artifact of a single metric.

4.5. Large-scale Validation

As a transfer test rather than another search, we compare conservative uniform scaling ($s = 1.25$), aggressive uniform scaling ($s = 2.50$), and fixed C3 (TCAS) on 300 objects.

Table 5 shows that $s = 1.25$ has weaker structural scores, while $s = 2.50$ obtains the highest Edge-SSIM but lower PSNR than C3. C3 reaches comparable FG-SSIM to $s = 2.50$, improves PSNR to 18.86, and slightly reduces FG-LPIPS, indicating that timestep-conditioned scaling preserves structural alignment while improving signal quality.

Table 5. Large-scale validation on 300 objects. Values are absolute metrics of generated outputs with respect to ground truth.

Method	FG-SSIM↑	Edge-SSIM↑	PSNR↑	FG-LPIPS↓
$s = 1.25$	0.430	0.513	17.95	0.174
$s = 2.50$	0.473	**0.559**	17.90	0.172
C3 (TCAS)	**0.476**	0.529	**18.86**	**0.171**

Table 6. Large-scale texture-diagnostic validation on 300 objects. Higher values indicate stronger foreground color variation or high-frequency response, but they are interpreted together with qualitative audits rather than as standalone texture-quality scores.

Method	Lap Var↑	RGB Std↑	Grad Mag↑
$s = 1.25$	**0.0151**	**0.0408**	**0.3258**
$s = 2.50$	0.0095	0.0251	0.2616
C3 (TCAS)	0.0129	0.0377	0.2919

Because PSNR is not a texture-quality metric, we further compute Laplacian variance, RGB standard deviation, and gradient magnitude on the same 300 objects as diagnostic evidence of variation loss. All results use the same objects, seed, 50 denoising steps, and adapter weights.

Table 6 shows that $s = 2.50$ has the lowest values across all three texture diagnostics, consistent with the flattening observed in Figure 4. C3 keeps comparable FG-SSIM while retaining higher Laplacian variance, RGB standard deviation, and gradient magnitude.

Table 7 shows that $s = 2.50$ retains only 67.8% Laplacian variance, 64.3% RGB standard deviation, and 80.7% gradient magnitude relative to $s = 1.25$. C3 retains 83.5%, 97.0%, and 88.3%, respectively, so it still incurs some reduction but substantially alleviates full high-scale flattening.

Object-level statistics show that C3 outperforms $s = 2.50$ on 87.7%, 95.0%, and 83.0% of objects in Laplacian variance, RGB standard deviation, and gradient magnitude, with bootstrap 95% confidence intervals not crossing zero. It also wins on 74.3% of objects in PSNR, while the FG-SSIM interval crosses zero, indicating no significant structural sacrifice.

Perceptual validation with CLIP-IQA. To address the concern that proxy texture statistics do not directly measure perceptual quality, we additionally evaluate CLIP-IQA [34], which builds on CLIP representations [35], on a 50-object validation subset as a complementary no-reference perceptual signal. CLIP-IQA is used as complementary perceptual evidence rather than a complete replacement for human preference evaluation.

Table 8 shows that C3 achieves a foreground CLIP-IQA score of 0.4906 versus 0.4860 for $s = 2.50$, winning on 94.0%

Table 7. Texture-diagnostic retention ratios and object-level stability of C3. Ratios are measured relative to $s = 1.25$; object-level win rates and confidence intervals for C3 over $s = 2.50$ are reported in the text. These ratios measure retained foreground variation rather than perceptual texture quality by themselves.

Method	Lap Var Ratio↑	RGB Std Ratio↑	Grad Mag Ratio↑
$s = 1.25$	**1.000**	**1.000**	**1.000**
$s = 2.50$	0.678	0.643	0.807
C3 (TCAS)	0.835	0.970	0.883

Table 8. CLIP-IQA perceptual validation on 50 validation objects. Scores are reported for foreground regions and full rendered views. Higher is better.

Method	CLIP-IQA (FG)↑	CLIP-IQA (Full)↑	Win vs. $s = 2.50$
$s = 1.25$	**0.4920 ± 0.0032**	**0.4918 ± 0.0030**	**98.0%**
$s = 2.50$	0.4860 ± 0.0018	0.4857 ± 0.0017	–
C3 (TCAS)	0.4906 ± 0.0028	0.4903 ± 0.0025	94.0%

of objects. The paired test is significant ($p < 10^{-6}$), with a 95% confidence interval of [0.0037, 0.0054]. The ordering $s = 1.25 >$ C3 $> s = 2.50$ supports the diagnosis that aggressive scaling reduces perceptual quality, while TCAS recovers most of this loss with stronger geometry than conservative scaling.

Blinded preference study. To further examine whether the diagnostic texture differences are perceptually meaningful, we conducted a blinded three-alternative forced-choice preference study on 30 validation objects. Twenty-four participants viewed anonymized results from conservative uniform scaling, aggressive uniform scaling, and C3 in randomized order, and selected the best result for texture naturalness, shape consistency, and overall quality. This yields 720 valid first-choice decisions for each criterion. The study complements the proxy diagnostics by testing whether the observed texture and structure differences are reflected in human preference.

Table 9. Blinded three-alternative first-choice preference study. Percentages are computed over 720 valid decisions per criterion.

Method	Texture naturalness↑	Shape consistency↑	Overall quality↑
$s = 1.25$	**332/720 (46.1%)**	96/720 (13.3%)	176/720 (24.4%)
$s = 2.50$	104/720 (14.4%)	**331/720 (46.0%)**	126/720 (17.5%)
C3 (TCAS)	284/720 (39.4%)	293/720 (40.7%)	**418/720 (58.1%)**

The preference pattern in Table 9 follows the intended trade-off. Conservative scaling is most often selected for texture naturalness, while aggressive scaling is most often selected for shape consistency. C3 stays close to both single-scale baselines on their respective strengths and receives the highest overall preference, with 58.1% of all overall-quality votes. Thus, TCAS is not claimed to maximize every individual criterion; it provides a perceptually preferred balance between geometric adherence and texture preservation.

Overall, the 300-object experiments show that TCAS maintains structural alignment while reducing the flattening effect of full high-scale scaling. C3 still has some diagnostic decrease relative to $s = 1.25$, so we do not claim that TCAS eliminates texture loss or guarantees superior texture in every case. Instead, the metric, CLIP-IQA, visual, and preference evidence jointly indicate that it reduces the high-scale over-control pattern and offers the stronger overall trade-off.

4.6. FAC Adaptive Correction Framework

TCAS is training-free and lightweight, but its residual scaling is uniform across spatial positions and frequency components. To instantiate the same principle in a learned adaptive form, we evaluate FAC on the 300-object validation set. FAC keeps the base model, VAE, and original GeoTex-Adapter frozen, adding LTAG timestep gating, GSG spatial gating, and FSC frequency-selective correction on top of TCAS. Because FAC learns additional parameters, TCAS remains the training-free main strategy.

Table 10. Ablation results of the FAC adaptive correction framework on the 300-object validation set. Full FAC adds spatial gating and frequency-selective correction on top of the TCAS temporal schedule.

Variant	PSNR↑	SSIM↑	FG-PSNR↑	FG-SSIM↑
TCAS (baseline)	18.86	0.8737	11.66	0.4755
LTAG only	17.87	0.8732	10.47	0.4635
LTAG + GSG	18.09	**0.8794**	10.43	0.4640
Full FAC	**19.28**	0.8793	**11.77**	**0.4830**

Table 10 reports controlled ablations. LTAG alone underperforms TCAS, so learned temporal gating alone cannot stably replace the fixed schedule. LTAG+GSG improves SSIM but remains lower in PSNR, FG-PSNR, and FG-SSIM. Full FAC performs best overall: PSNR rises from 18.86 to 19.28, FG-PSNR from 11.66 to 11.77, and FG-SSIM from 0.4755 to 0.4830.

Object-level statistics further show that Full FAC outperforms TCAS on 81.3% and 72.0% of objects in PSNR and FG-SSIM, respectively, and paired significance tests for both metrics are significant ($p < 0.0001$). This indicates that the gains of FAC are stable across most objects rather than being driven by a small number of samples.

Thus, TCAS remains the main strategy when no additional training is desired, while FAC shows that stage-wise adapter control can generalize to learned spatial and frequency correction.

5. Limitations

This paper focuses on inference-time adapter scaling and does not retrain the base multi-view diffusion model or adapter. TCAS can alleviate high-scale texture flattening, but cannot fundamentally solve invisible-region completion, complex material synthesis, or cross-view detail consistency.

Second, Laplacian variance, RGB standard deviation, and gradient magnitude are proxy metrics for texture degradation. They detect high-frequency loss and surface smoothing, but noise, erroneous patterns, or artifacts can also increase them. We therefore interpret them with local visual audits, CLIP-IQA validation, and a small blinded preference study. A more complete evaluation should add larger-scale blinded user studies, full-reference metrics such as DISTS [36], and FID/KID on larger rendered-view test sets.

In addition, TCAS uses fixed stage boundaries and C3 = (1.25, 2.50, 1.25). It is a validated strategy for the studied geometry-adapter pipeline rather than a universally optimal schedule, and has not yet been adapted to object complexity, material type, reference quality, or sampler families.

Finally, FAC requires additional training and therefore does not replace the plug-and-play property of TCAS; it is treated as a learned adaptive instantiation of the same principle.

6. Conclusion

This paper analyzes adapter scaling in multi-view diffusion texture generation. Higher adapter scales improve structural metrics such as FG-SSIM, but this can mask texture flattening: in the 26-object audit, 92% of objects show smoother surfaces and weakened coherent local variation under $s = 2.50$. This reveals an adapter-specific temporal conflict not captured by treating scale as generic guidance strength.

We propose TCAS, whose C3 configuration applies strong adapter correction in the middle denoising stage and conservative scales in early/late stages. C3 is selected only on a 24-object probe set from 17 variants, then frozen. Comparisons with linear decay, cosine decay, linear warm-up, and smooth cosine-bump schedules show that common guidance-style schedules do not provide the same trade-off. On 300 objects, C3 achieves FG-SSIM comparable to $s = 2.50$ while improving PSNR by 0.96 dB.

The core contribution is therefore to reveal and validate a hidden shape-texture trade-off caused by geometry-adapter residual injection, and to show that a simple non-monotonic inference schedule mitigates flattening better than fixed high-scale control and the tested generic schedules. Because the texture diagnostics are proxies, we further verify the conclusion with CLIP-IQA, where C3 outperforms $s = 2.50$ on 94.0% of the 50-object subset, and with a blinded preference study where C3 receives 58.1% of overall first-choice votes. Adapter strength should thus be treated as a stage-dependent inference control variable rather than a single fixed hyperparameter.

FAC further shows that stage-wise adapter control can generalize from a fixed inference schedule to lightweight learned spatial and frequency correction.

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